

Scope: Gummel decoupled-Newton fallback for the steady-state driver

Date: 2026-06-22 **Status:** SCOPE + SCAFFOLD BUILT (M1+M2); M3/M4 OPEN **Owner area:** perovskite_sim/experiments/steady_state.py, physics/continuity.py, physics/interface_plane.py

0. Empirical update (2026-06-22) — scaffold built, basic Gummel STALLS

The M1+M2 scaffold is implemented (dormant, default-bit-identical): - `assemble_rhs(phi_frozen=...) + _residual_fn(phi_frozen=...)` — the frozen-phi carrier residual seam (`solver/mol.py`, `steady_state.py`). - `_gummel_point` — the decoupled outer loop (carrier Newton at frozen phi + `_qfl_poisson_relax` Poisson half + ln-space under-relaxation). Env-gated `SOLARLAB_SS_GUMMEL`; NOT wired into `run_jv_sweep_ss/solve_voc_ss` yet.

Measured on the deep-CBO point ($\Delta E_C = -1.0$, where the coupled Newton goes singular): the loop reduces the seed residual by ~ 13 orders ($4e13 \rightarrow O(1-1e9)$) but then **STALLS at a fixed point above the certification guard** — `res` ~ 3.6 with `iface_states`, $\sim 2e9$ without, and **identical at carrier_max_it 20 and 40**, i.e. a genuine *decoupled-Gummel stall*, not slow convergence. **This confirms the Section 9 primary risk: the basic decoupled scheme stalls at strong coupling.** So M1+M2 alone do NOT close the deep-CBO gaps. Closing them is the **M4 acceleration** work (the schedule risk): Anderson acceleration, a coupled-Newton polish that survives the singular Jacobian, or a transient-assisted Gummel. Until then the deep-CBO SS points stay documented as gapped (the transient covers that regime; `SS` $\sim$ transient there). The `phi_frozen` seam is the reusable, bit-identical foundation M4 builds on.

1. Motivation

The steady-state interface-states driver (`run_jv_sweep_ss(iface_states=True)`, `solve_voc_ss`) is the configuration that recovers the SCAPS interface-doping directions (`Nd_ETL` rising, etc.). It has one structural weakness: the coupled damped-Newton (`solve_steady_state`) **fails to converge in collapsed-junction and extreme-band-offset regimes**, where it falls back to the certified transient point-fallback, which then grinds for minutes per point or fails.

Two concrete symptoms motivated this scope:

1. **Deep-CBO SS gaps (the trigger).** On the ETL/PVK conduction-band-offset sweep, the SS series converges only over the flat-band region ($\Delta E_C \geq -0.16$ eV). The deep-offset points ($\Delta E_C \leq -0.2$ eV, where `V_oc` collapses to $\sim 0.33-0.83$ V) **do not converge in SS** even at a 300 s cap — yet the **transient solves them in ~ 8 s** and finds a real `J=0` crossing. So a crossing *exists*; the SS solver simply cannot reach it.
2. **The `Nd_ETL` $\leq 1e10$ xfail** (`tests/unit/experiments/test_steady_state.py` near-insulating gate) — see the critical caveat in Section 7; this is a *different* problem and is **not** fully solved by Gummel.

The de-spike + interface-channel calibration work (2026-06) made the operative regime fast and accurate; this scope is about the *hard tail* the coupled Newton can't reach.

2. Root cause

Every `assemble_rhs` call solves Poisson globally (`solver/mol.py:1693`), so `phi[i]` depends on `rho` at all nodes, and the Scharfetter-Gummel carrier fluxes depend on `phi`. The carrier Jacobian therefore acquires a **dense, phi-mediated tail** (measured: ~ 13.5 % of the Jacobian Frobenius weight outside a 2-node band). In the near-insulating / collapsed-junction regime the dielectric-relaxation mode makes that coupling extreme, and the dense FD Newton goes **singular** (the existing ridge fallback + transient assists are the symptom-management). The full-thermal-velocity interface-plane-state block (`iface_state_v_th = 1e5`) adds a second stiff mode; today both share **one global dense Jacobian**, so they compound.

Gummel decoupling breaks the dense phi-coupling: solve the carrier-transport equations at *frozen* phi (block-tridiagonal, well-conditioned), then relax phi at *frozen* quasi-Fermi levels (the analytic, unconditionally

well-posed dielectric step), and iterate. The two stiff modes are then handled by the operator each is well-conditioned for, instead of one ill-conditioned Newton.

3. The design

A Gummel **point solver** that replaces a single failed `solve_steady_state` point — used as an *opt-in fallback* before the transient fallback:

```
y = ensure_iface_block(y_seed, mat)
y = _qfl_poisson_relax(x, mat, y, V_app)      # Poisson half (EXISTS)
for it in range(max_outer):
    # --- carrier half: continuity at FROZEN phi ---
    phi = solve_poisson_once(y)               # one solve, then held fixed
    y = carrier_solve_frozen_phi(x, stack, mat, y, V_app, phi)  # NEW (~80-150 lines)
    y = underrelax_ln(y_old, y, omega)        # ln-space, positivity-free
    # --- interface-plane states: local per-interface subsystems ---
    if mat.N_iface_state: y = solve_iface_states_local(x, mat, y, phi)  # NEW (reuse solve_plane_densities)
    # --- Poisson half (EXISTS) ---
    y = _qfl_poisson_relax(x, mat, y, V_app)
    # --- convergence on the SAME coupled residual solve_steady_state uses ---
    res = peak_density_scaled_residual(assemble_rhs(0, y, ...))
    if res < tol: break
    if res rose: omega = max(0.1, 0.5*omega)  # adaptive damping (load-bearing)
if res > res_guard: raise SteadyStateError(...) # fail-loud, same contract
return y, res
```

Three pieces, two of which already exist:

Piece	Status	Source
Poisson half (nonlinear Poisson at frozen QFL, banded, damped)	EXISTS	<code>_qfl_poisson_relax</code> <code>steady_state.py:451-512</code>
Carrier half (continuity at frozen phi, block-tridiagonal banded solve)	NEW (~80-150 lines)	mirrors the Poisson half; <code>carrier_continuity_rhs</code> is already phi-pure
Interface-plane states as local per-interface subsystems	NEW (small)	template <code>solve_plane_densities</code> <code>interface_plane.py:385-433</code>

4. Why it is tractable (verified from source)

- **carrier_continuity_rhs** is already **phi-pure** — it takes `phi` as an argument and never solves Poisson (`continuity.py:101`, reads `phi` only at 132-139). A frozen-phi carrier residual is a direct call (zero edits) or a `phi=` kwarg on `assemble_rhs` (~5-10 lines wrapping `mol.py:1670-1695`).
- The **carrier Jacobian at frozen phi** is **block-tridiagonal** — SG flux is nearest-neighbour (`fe_operators.py:58-68`), divergence couples node `i` to `i+1` only (`continuity.py:244`), recombination is node-local (`recombination.py:47`). So it solves with `scipy.linalg.solve_banded`, exactly as the Poisson half already does (`steady_state.py:496`).
- The **interface-plane state block** is **block-diagonal per interface** — each contributor writes only into its own `base=4k` slot (`interface_plane.py:213,473,577`; `mol.py:1885`), no inter-interface coupling. With `phi` + carriers frozen, each interface is a closed 4-unknown system solvable with the existing damped log-Newton (`solve_plane_densities`, `interface_plane.py:385-433`). The only bulk back-coupling is the eval-node drain (`mol.py:1907-1910`), folded as a lagged source into the next carrier sweep. (Today this block lives in the *global* dense Jacobian, `steady_state.py:298,344-353` — decoupling it is itself a win.)

5. Implementation plan (milestones)

M	Deliverable	Est.
M1	Frozen-phi carrier-continuity residual + banded carrier Jacobian (<code>carrier_solve_frozen_phi</code>); unit test vs <code>assemble_rhs</code> carrier rows at a converged state (must match).	1 d
M2	Gummel outer loop (<code>_gummel_point_fallback</code>) alternating M1 + <code>_qfl_poisson_relax</code> , adaptive ln-space under-relaxation, certify-or-raise on the peak-density residual. Wire as opt-in fallback before <code>_transient_point_fallback</code> at the two except <code>SteadyStateError</code> sites (<code>run_jv_sweep_ss:602-610</code> , <code>solve_voc_ss:657-661</code>); gate <code>SOLARLAB_SS_GUMMEL</code> (default OFF) + <code>gummel_fallback=False</code> kwarg.	1 d
M3	Interface-plane states as local per-interface subsystems inside the loop (reuse <code>solve_plane_densities</code>); validate on the deep-CBO points that currently gap ($\Delta E_C \leq -0.2$).	1-1.5 d
M4	Deep-regime convergence hardening (damping schedule, omega floor, semi-implicit bulk drain) + the test surface in Section 8 + docs.	1.5-2 d

Core deliverable (M1-M4): converge the deep-CBO / collapsed-but-bracketed SS points -> ~4-6 days. This fills the CBO SS gaps (the user's ask).

6. Integration & safety (bit-identical default)

- The Gummel helper is reached **only after the coupled Newton already raised `SteadyStateError`** — every point that converges today never enters it. Even with the flag on, converging points stay **bit-identical**; only the currently-*failing* regime changes behaviour. Zero edits to the Newton loop.
- Default OFF via the established opt-in pattern (cf. `SOLARLAB_SS_JAC_REUSE`, `te_softness`, `iface_states`). On Gummel failure, **fall through** to `_transient_point_fallback` so existing behaviour is strictly preserved.
- Fail-loud contract kept: certify on the same peak-density residual scale (`steady_state.py:546-553`), raise on failure — no silent fallback.

7. CRITICAL caveat — Gummel does NOT flip the `Nd_ETL <= 1e10` xfail

The near-insulating xfail (`test_steady_state.py` low-doping gate) is a **physics/boundary-condition problem, not a convergence problem**. The voltage walk already *completes* there and finds **no $J=0$ crossing below 1.6 V** — both the SS driver and the transient flat-band result agree there is no crossing (the model's crossing sits near 1.29 V, above the detailed-balance ceiling). SCAPS reports $V_{oc} \sim 1.10$ V there only via its near-insulating **contact convention** (flat-band finite-S contact), which SolarLab does not apply by default.

Therefore: - Gummel gives **convergence** in the near-insulating regime (kills the dielectric-relaxation stiffness the coupled Newton chokes on) — necessary but **not sufficient**. - Flipping the xfail to xpass *additionally* requires a **near-insulating contact boundary condition** (the existing `flat_band_contacts` path is the candidate seam) so that a physical crossing exists to be found. - That combined effort is **~10-18 days with a go/no-go** after the first crossing attempt, and is **out of scope** for the core deliverable. The CBO gaps (Section 1.1) do *not* depend on it — they have a real crossing already.

8. Acceptance criteria / test surface

Core (M1-M4): 1. **Bit-identical default** — full unit + the SS parity suite (`test_ss_jv_matches_frozen_ion_transient`, `test_direct_voc_consistent_with_jv`) unchanged with the flag OFF, and unchanged for every point that converges today with the flag ON. 2. **M1 correctness** — `carrier_solve_frozen_phi` reproduces `assemble_rhs` carrier rows at a converged state to tolerance. 3. **Deep-CBO convergence (the headline)** — `run_jv_sweep_ss(gummel_fallback=True)` converges the delta $E_C \leq -0.2$ eV CBO points that currently gap, with FOM ($V_{oc}/J_{sc}/FF/PCE$) physical and the SS curve overlaying the transient there (CBO is band-offset-limited). 4. **Fail-loud preserved** — a genuinely unsolvable point still raises `SteadyStateError`.

Stretch (separate effort, Section 7): 5. `Nd_ETL <= 1e10` flips to xpass *only* in combination with the contact-BC change — tracked as its own task with a go/no-go.

9. Risks & mitigations

Risk	Mitigation
Gummel is only linearly convergent; stalls/oscillates at high injection (V_{oc} knee, large $n \cdot p$)	Adaptive ln-space under-relaxation (ω start 1, halve on residual rise, floor ~ 0.1); fall through to the transient fallback on stall.
n, p form a 2×2 node block (recombination cross-term)	Iterate the node-local $n-p$ coupling, or use a 2-field banded solve.
Interface bulk-drain ($v_{th}=1e5$) is large + lagged $\rightarrow$ oscillation	Semi-implicit / under-relaxed drain; keep iface-states local-solve inside the outer loop.
Frozen-phi supply exponential clamped at 30 $\rightarrow$ deep CBO can sit on the clamp and lose sensitivity	Re-project phi each outer sweep (not once); validate CBO sensitivity is retained.
Partial (non-converged) Gummel sweeps would shift validated FOM	Certify-or-raise on the coupled residual; never return a non-converged point.
OneDrive sync can overwrite an edited solver file before <code>git add</code>	Stage promptly (known repo gotcha).

10. Scope boundaries

- **In:** a Gummel point-solver fallback that converges collapsed-junction / deep-offset SS points where the coupled Newton fails (fills the CBO SS gaps); opt-in, default-off, bit-identical; interface-states handled as local subsystems.
- **Out:** the near-insulating contact-convention / `Nd_ETL <= 1e10` physics fix (separate task, Section 7); making Gummel the *default* SS path (it is a fallback only); any change to the coupled-Newton hot loop.